

Bridging Tech Expertise to New Hires Ui Path

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Retaining Knowledge

This workflow was developed to address the challenge of retaining knowledge in manufacturing environments. Over time, machines are patched and modified, and the understanding of these changes exists primarily with experienced technicians.

When these individuals are unavailable or leave their roles, that knowledge is often lost, creating gaps for new hires who rely more on training than real-world experience. This workflow captures and preserves that experience, making it accessible to support consistent and efficient operations.

Problem

In many manufacturing facilities, older machines are patched, modified, and adjusted over time to reduce downtime and control costs. These workarounds often exist only in the knowledge of experienced technicians.

When these experts leave or are unavailable, their practical knowledge is lost. Manuals and standard procedures rarely reflect the reality of the machines, leaving new technicians without guidance for both common and rare issues. This knowledge gap can lead to inefficiencies, increased downtime, and safety risks.

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Problem

Key Points:

- **Loss of expert knowledge:** Critical workarounds leave with experienced technicians.
- **Equipment complexity:** Informal modifications are undocumented.
- **Rare but critical issues:** New staff may never encounter them until it's urgent.
- **Increased downtime:** Troubleshooting without guidance slows production.
- **Higher maintenance costs:** Trial-and-error replaces proven fixes.
- **Safety risks:** Infrequent hazards may be overlooked.
- **Knowledge decay:** Each departure erodes practical operational knowledge.

Traditional Knowledge Retention Workflow

1. **Manual Interviews and Notes**

- HR or supervisors conduct periodic interviews with experienced technicians.
- Notes are taken manually on paper or in shared documents.

2. **Limited Documentation**

- Workarounds and non-standard machine fixes may be recorded in text files, Excel sheets, or internal manuals.
- Often inconsistent and depends heavily on the individual recording it.

3. **Ad hoc Knowledge Sharing**

- Experienced technicians share tips verbally or during shadowing sessions.
- New hires learn mostly by observation, trial-and-error, or instruction from senior staff.

4. **Review and Approval**

- Management reviews documented knowledge periodically, if at all.
- No standardized method to flag critical vs. non-critical issues.

5. **Updating Process Documents**

- Process documents or SOPs are updated only when major issues occur.
- Non-standard equipment behavior is often missed or only partially captured.

6. **Limited Accessibility**

- Information is scattered across files, emails, or binders.
- Difficult for HR, new technicians, or maintenance staff to access quickly.

Traditional Knowledge Retention Workflow

Key Limitations of the Traditional Workflow:

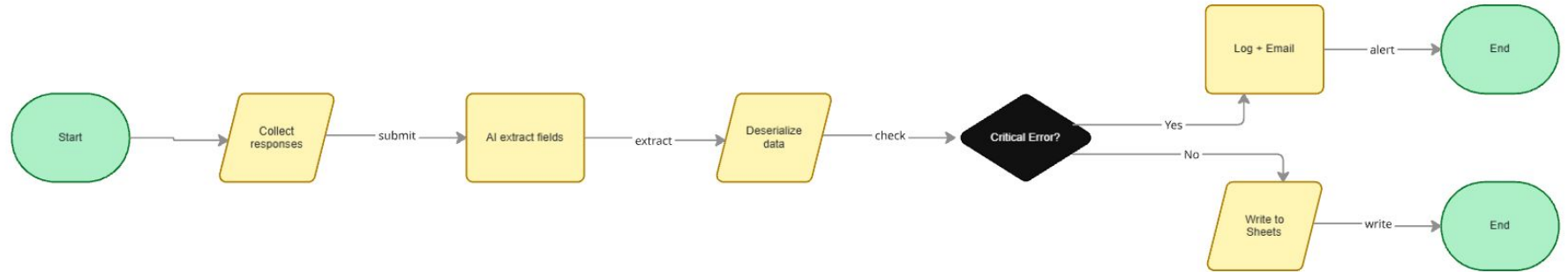
- Relies heavily on memory and manual effort.
- Critical knowledge may be lost if staff leave or retire.
- Time-consuming for HR and management.
- Inconsistent documentation and delayed updates.
- Limited traceability for rare or high-risk events.

Solution

To address the challenge of knowledge loss and infrequent but critical machine issues, this workflow captures and preserves the expertise of experienced technicians. While formal education provides a baseline understanding of machinery, it is **hands-on experience**—years of working with specific machines, patches, and unique workarounds—that enables technicians to handle unusual failures effectively.

This workflow ensures that the **practical, experience-driven knowledge** is retained and made accessible, bridging the gap between theoretical training and real-world problem-solving.

RPA workflow diagram



RPA workflow diagram

Collect Technician Responses

- HR or staff gather responses via interviews, logs, or text-to-speech.
- Include workarounds, maintenance insights, safety hazards, and rare issues.

AI Extraction of Key Fields

- Extract machine name, error code, standard fix, experienced fix/workaround, safety concerns, severity (Low, Medium, Critical), and estimated fix time.

Deserialize Extracted Data

- Convert AI output into usable variables for workflow processing.
- Ensure all fields are correctly mapped.

RPA workflow diagram

Handle Critical Responses

- If severity is **Critical**:
 - Log for traceability.
 - Send email to management for review.
 - Workflow ends; not written to Sheets.

Handle Non-Critical Responses

- If severity is **Low or Medium**:
 - Automatically write to Google Sheets.
 - Include all extracted fields.

Workflow Ends

- Non-critical responses stored in Sheets for new hires.
- Critical responses await management review

RPA workflow (tech reponses)

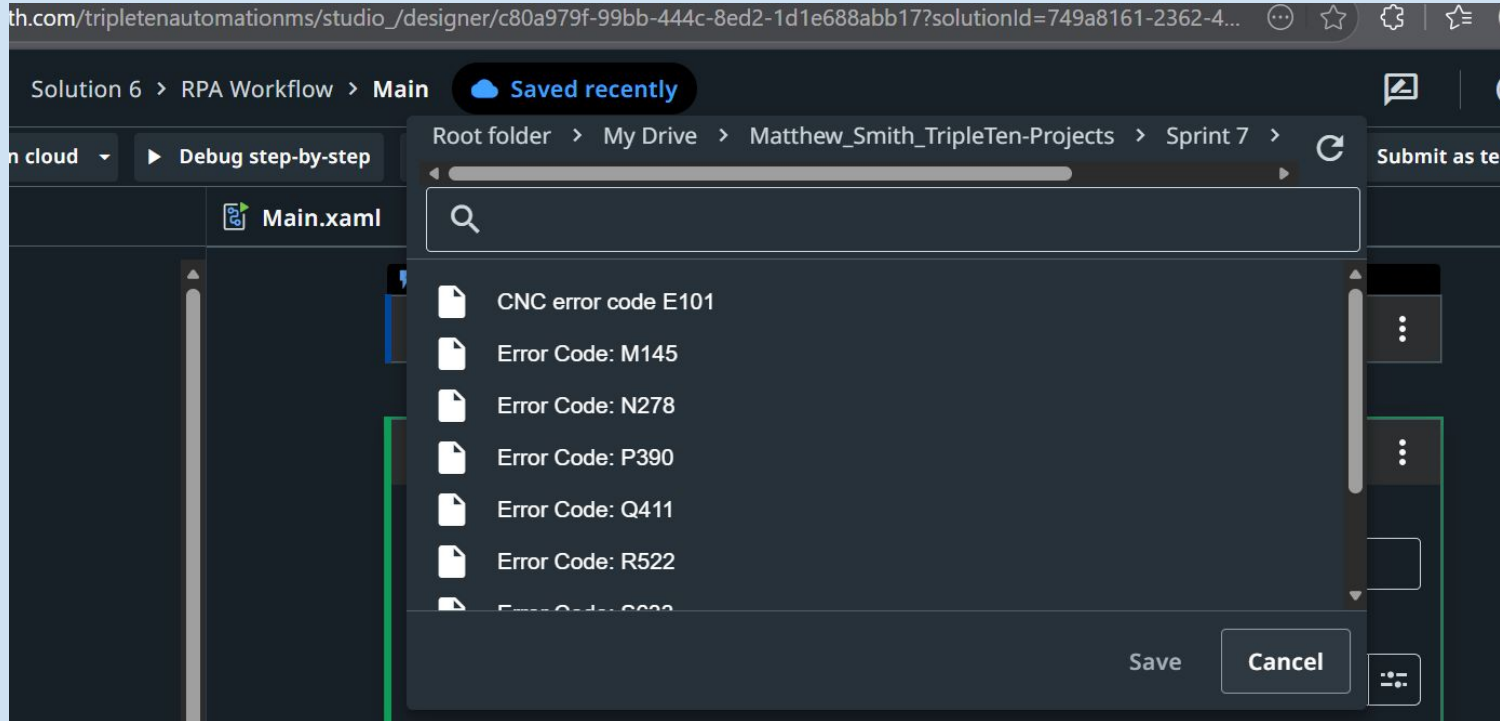
Full technician responses—collected via HR interviews, written logs, or text-to-speech AI—capture **informal workarounds, context, and rare scenarios** that rigid questions miss.

They allow technicians to **mentally walk through situations uninterrupted**, helping them think through problems clearly.

AI can extract all key elements—standard fix, workaround, safety concerns, severity, and estimated fix time—from one response.

This preserves **tacit knowledge**, reduces gaps when experienced staff leave, and gives new hires deeper operational insight.

RPA workflow (tech reponses)



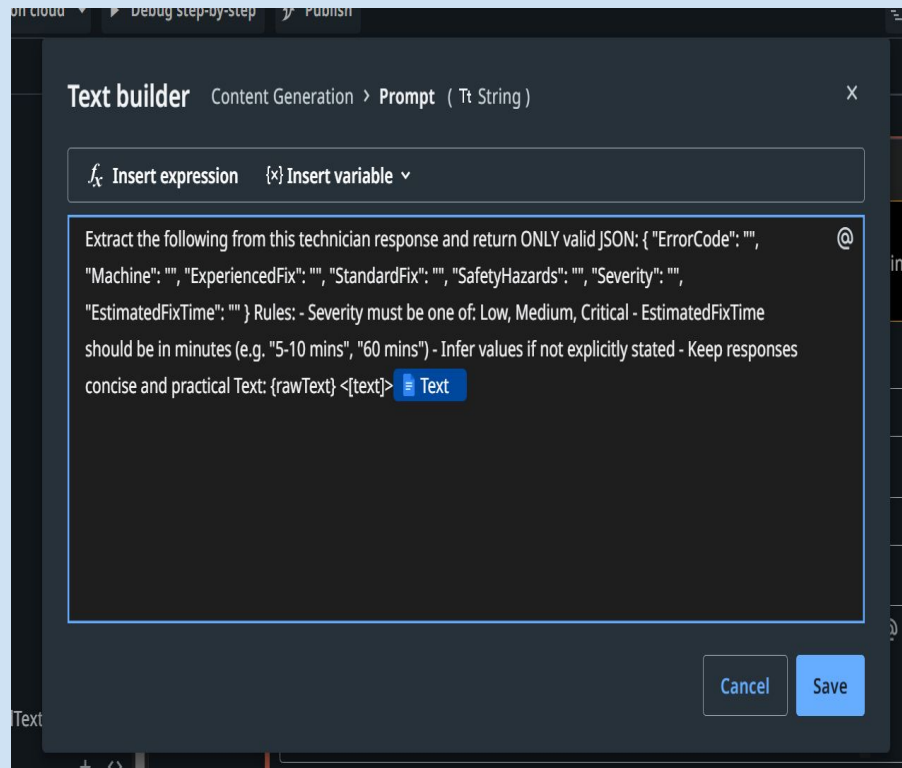
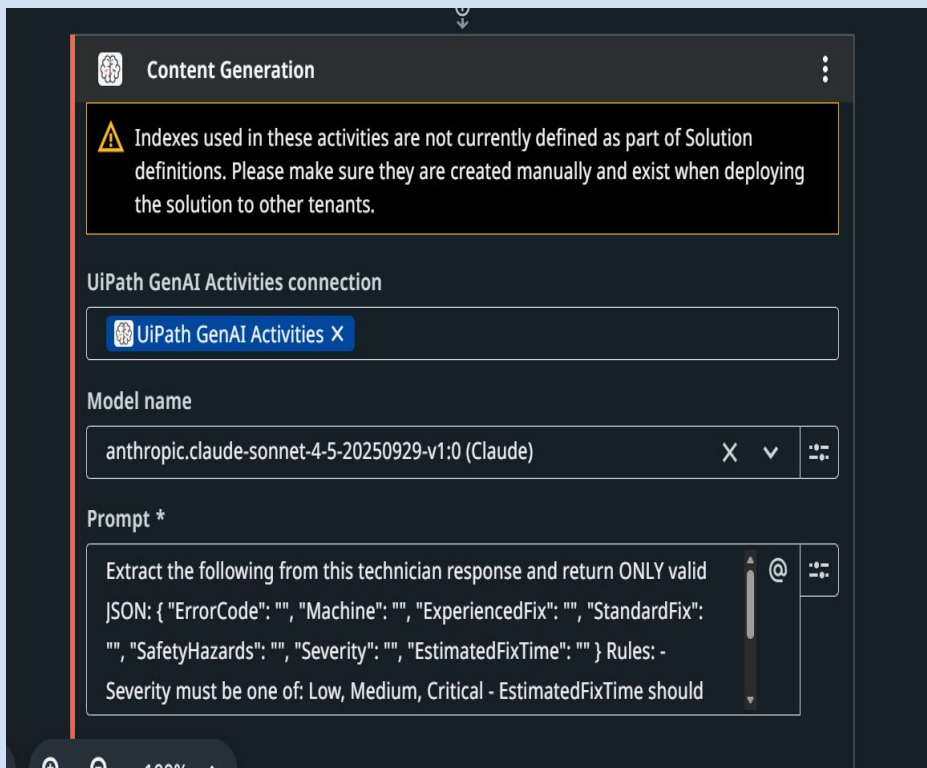
RPA workflow (retrieve)

The screenshot displays an RPA workflow editor with a dark theme. The workflow consists of two steps:

- Manual Trigger**: The first step, featuring a person icon and a menu icon.
- Read Text**: The second step, featuring a document icon and a menu icon. It is configured with the following properties:
 - Google Docs connection**: A text field containing the email address `ms1479.ms@gmail.com` with a close button (X).
 - Document***: A text field containing the text `Error Code: M145`, accompanied by a folder icon and a list icon.
 - Text to read***: A dropdown menu currently set to `All text`.
 - Show additional properties**: A link with a downward arrow to expand the configuration options.

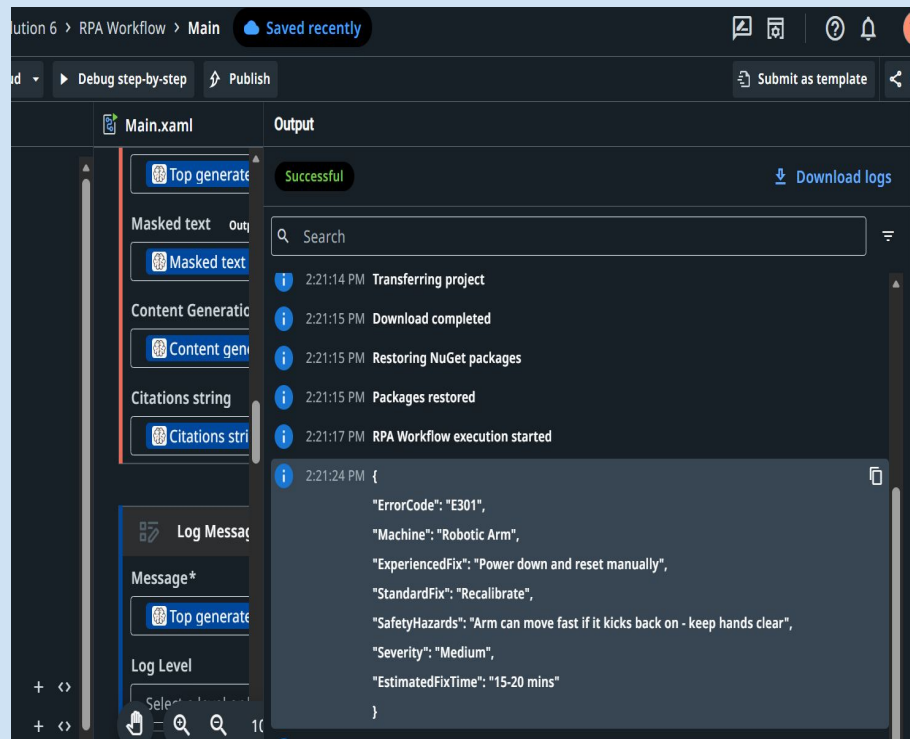
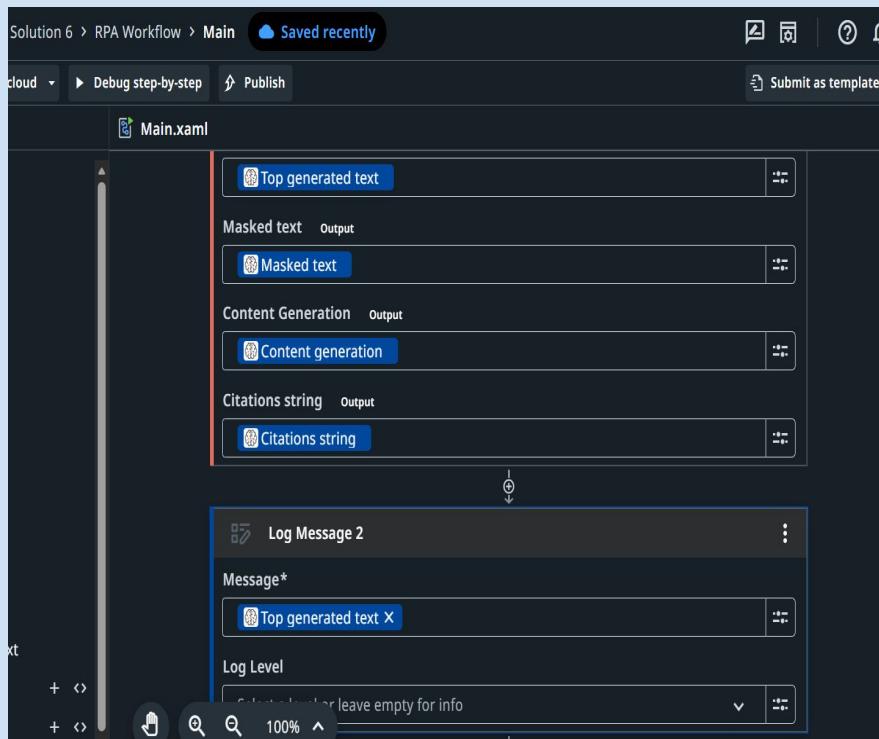
Flow connectors (downward arrows with a plus sign) are visible between the steps and at the bottom of the workflow area.

RPA workflow (extract)



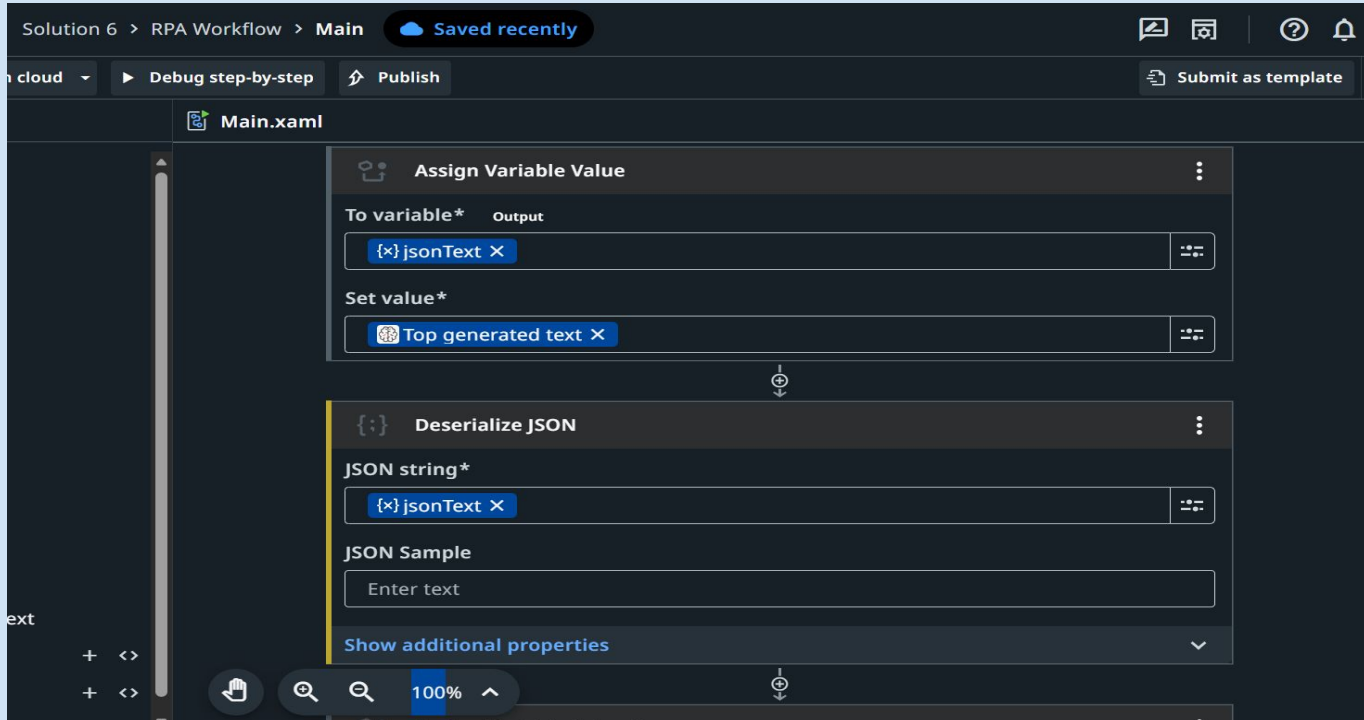
RPA workflow (extract)

AI outputs in JSON



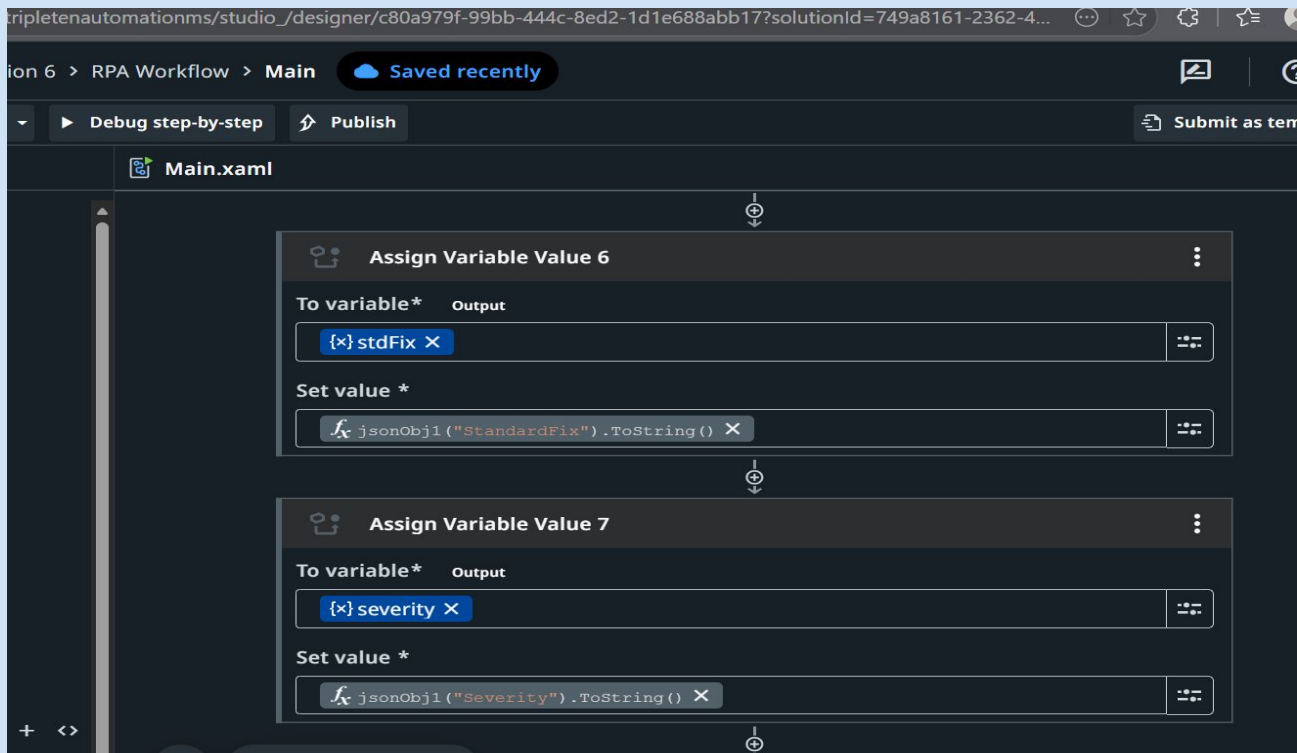
RPA workflow (deserialize)

Deserialization turns AI-extracted data into usable variables for the workflow.



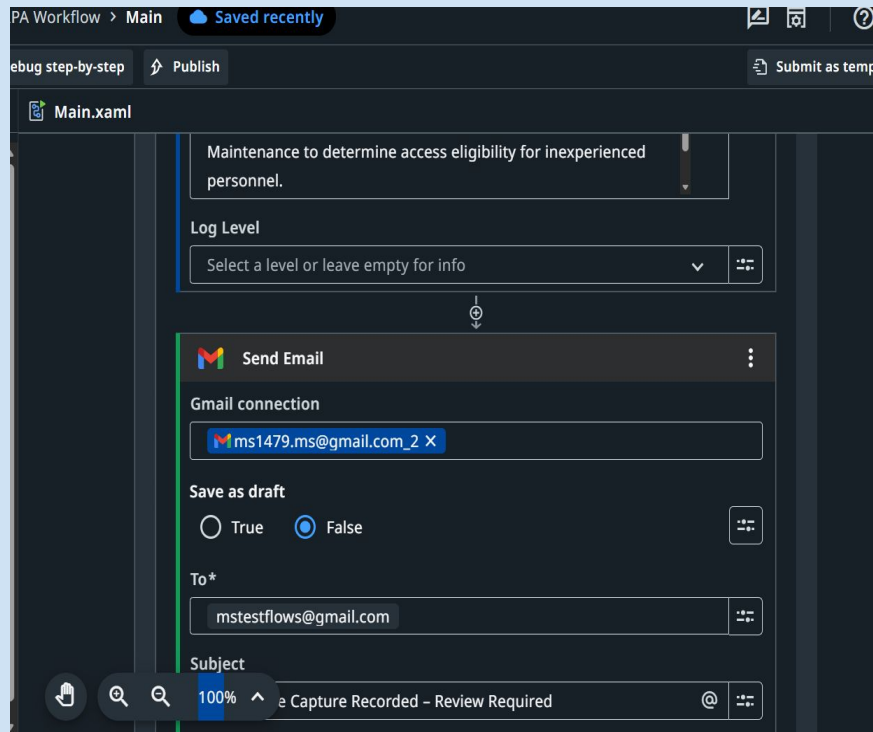
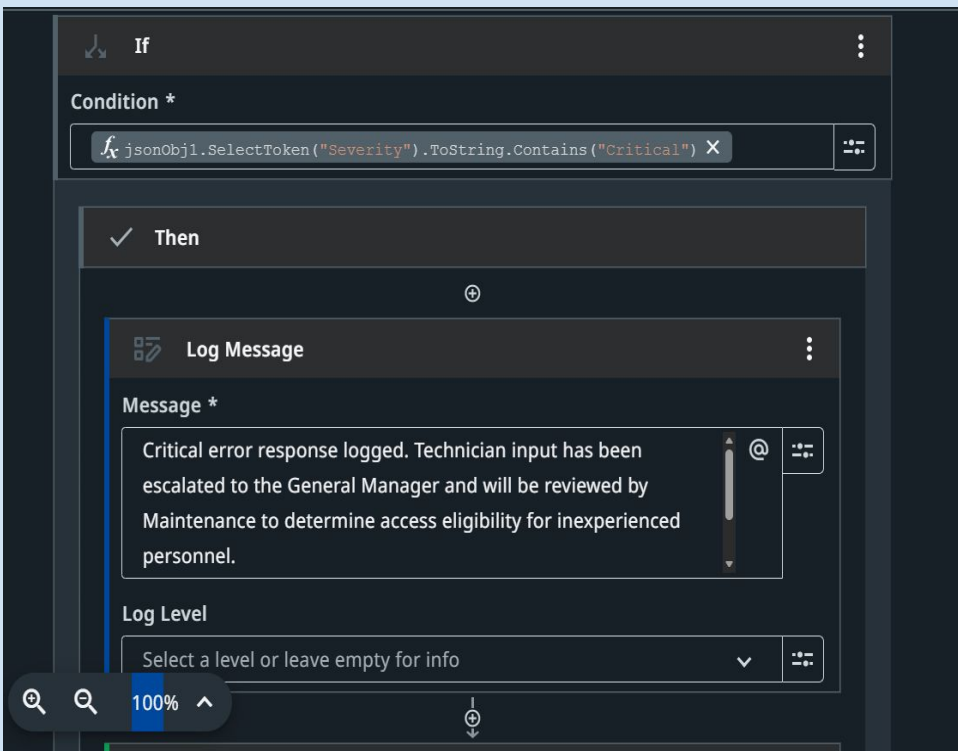
RPA workflow (variables)

Variables let the workflow use each deserialized field for processing.



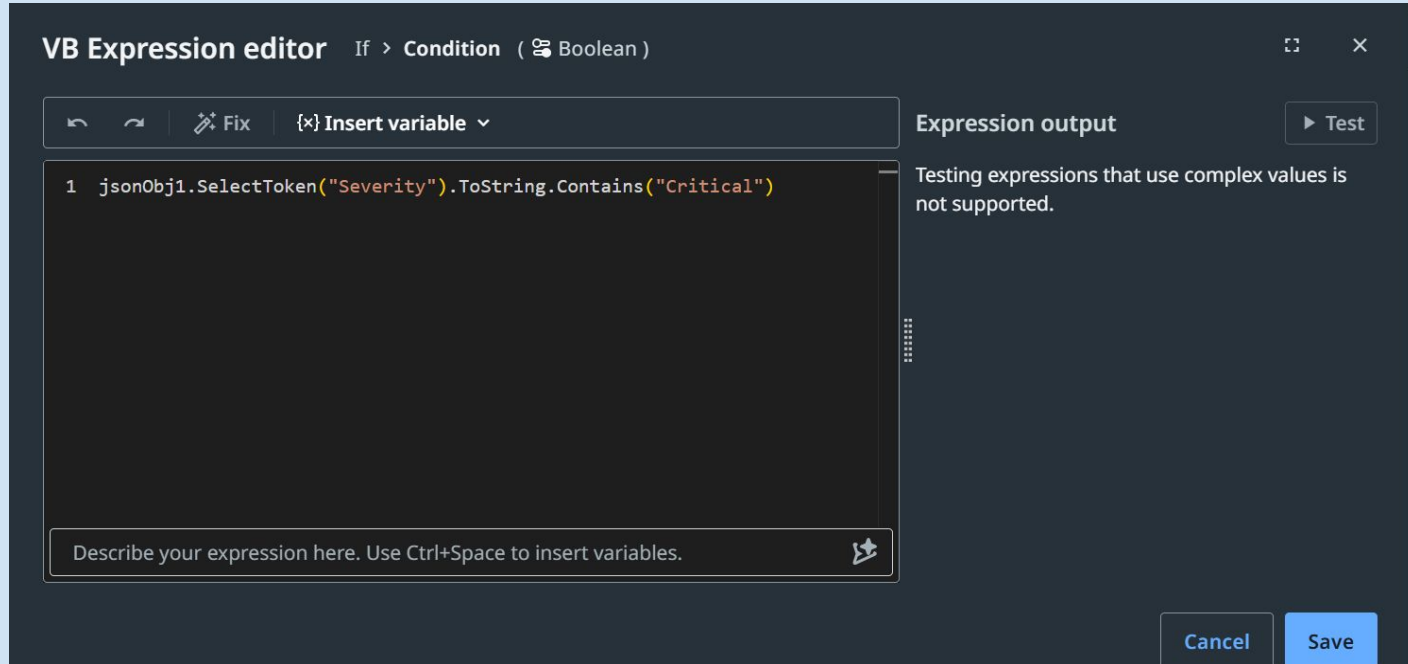
RPA workflow (decision)

“The If condition checks the response severity to decide whether to email management or write to Sheets.”



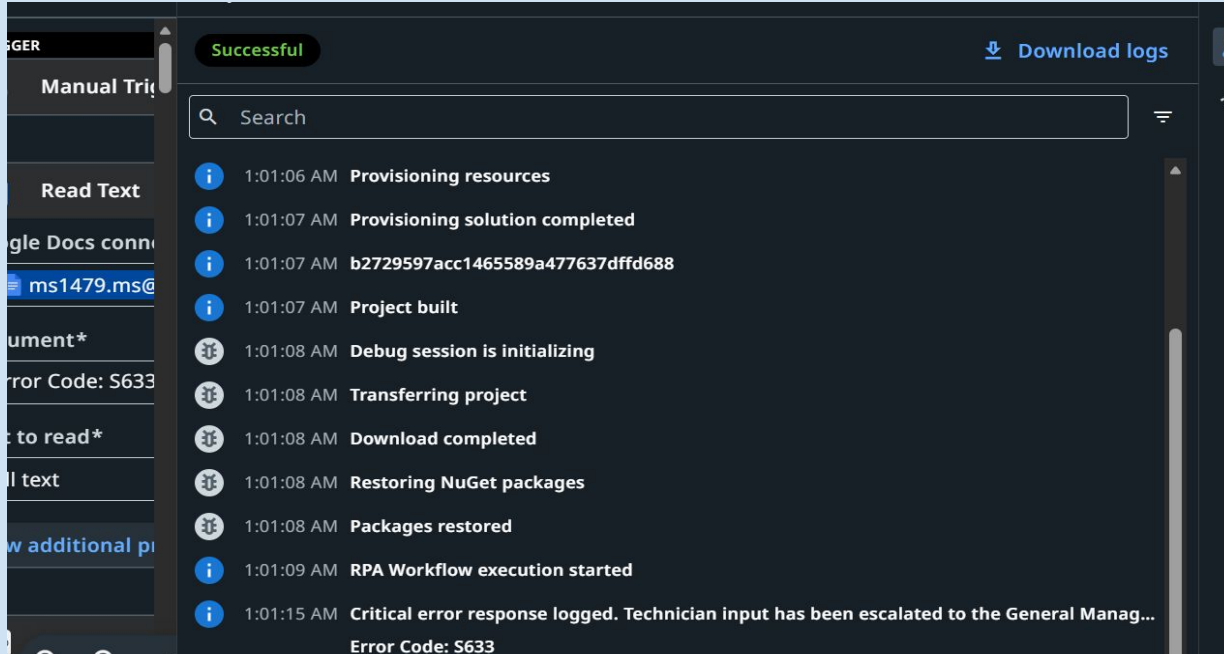
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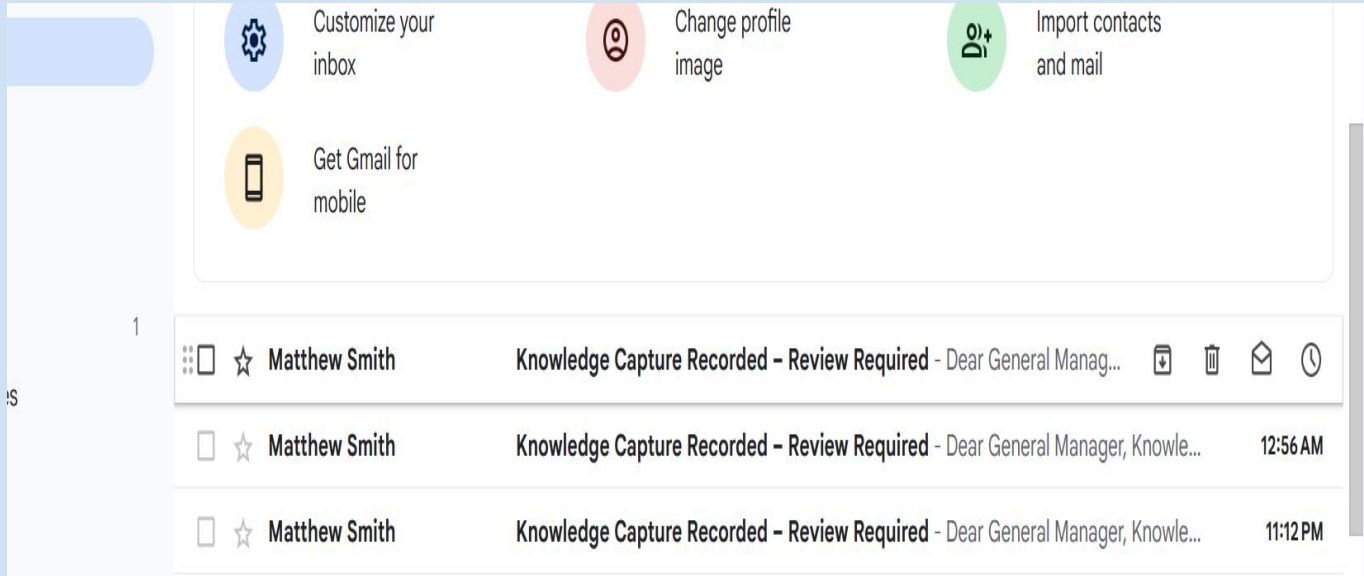
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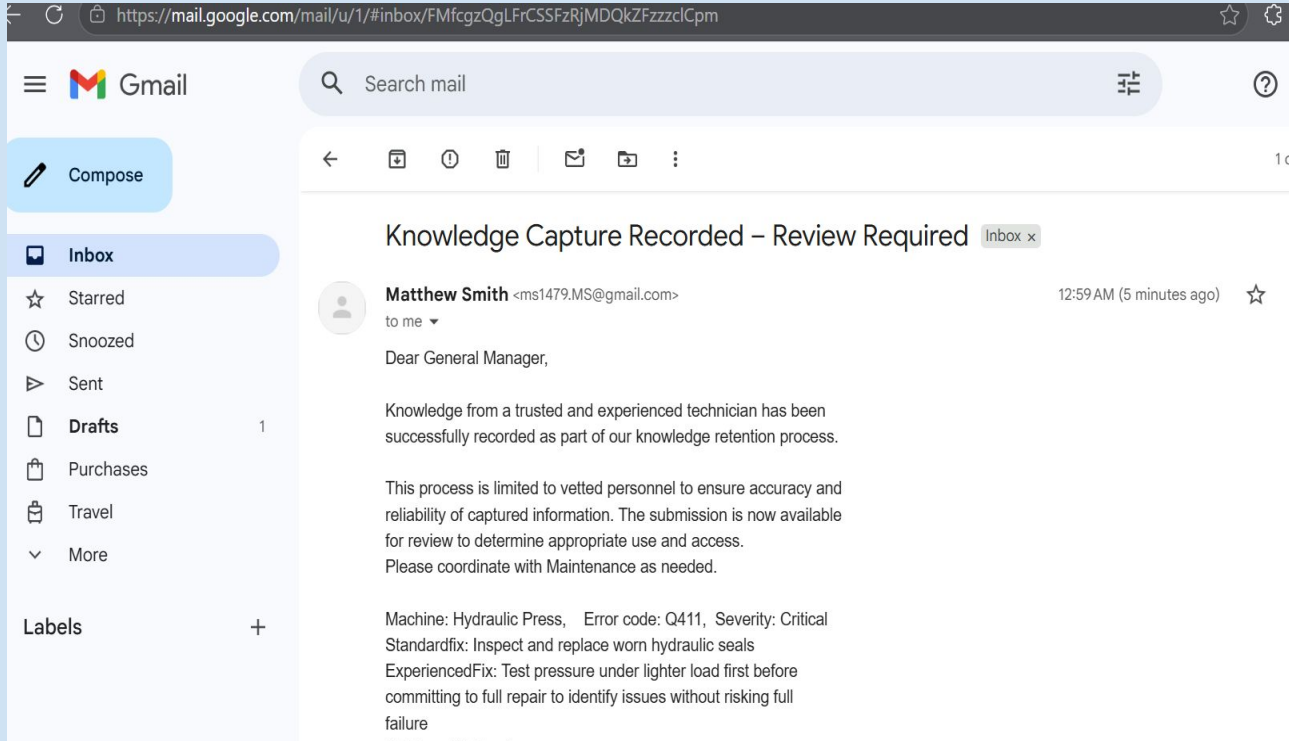
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“The If condition checks the response severity to decide whether to email management or write to Sheets if critical creates a log.”

The screenshot shows a 'Text builder' window with a breadcrumb trail 'Send Email > Body (Text String)'. At the top, there are two buttons: 'Insert expression' (with a function symbol) and 'Insert variable' (with a variable symbol and a dropdown arrow). The main text area contains the following content:

Dear General Manager,

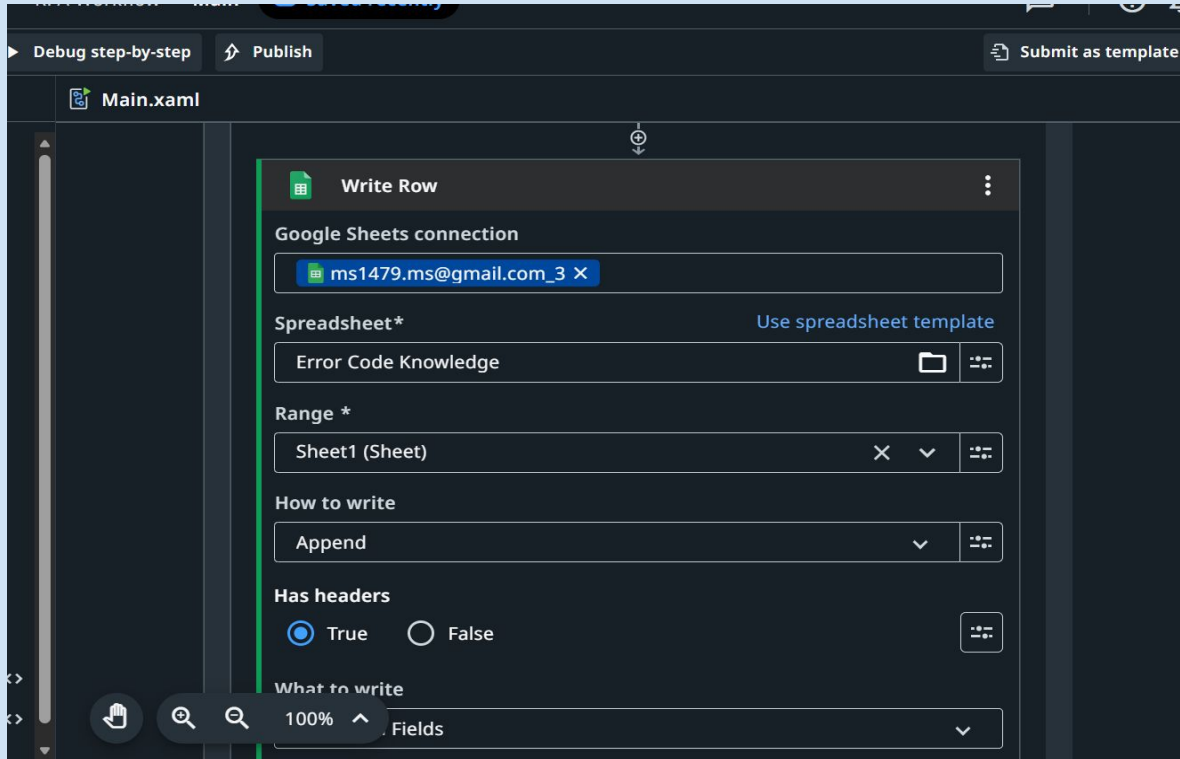
Knowledge from a trusted and experienced technician has been successfully recorded as part of our knowledge retention process.

This process is limited to vetted personnel to ensure accuracy and reliability of captured information. The submission is now available for review to determine appropriate use and access. Please coordinate with Maintenance as needed.

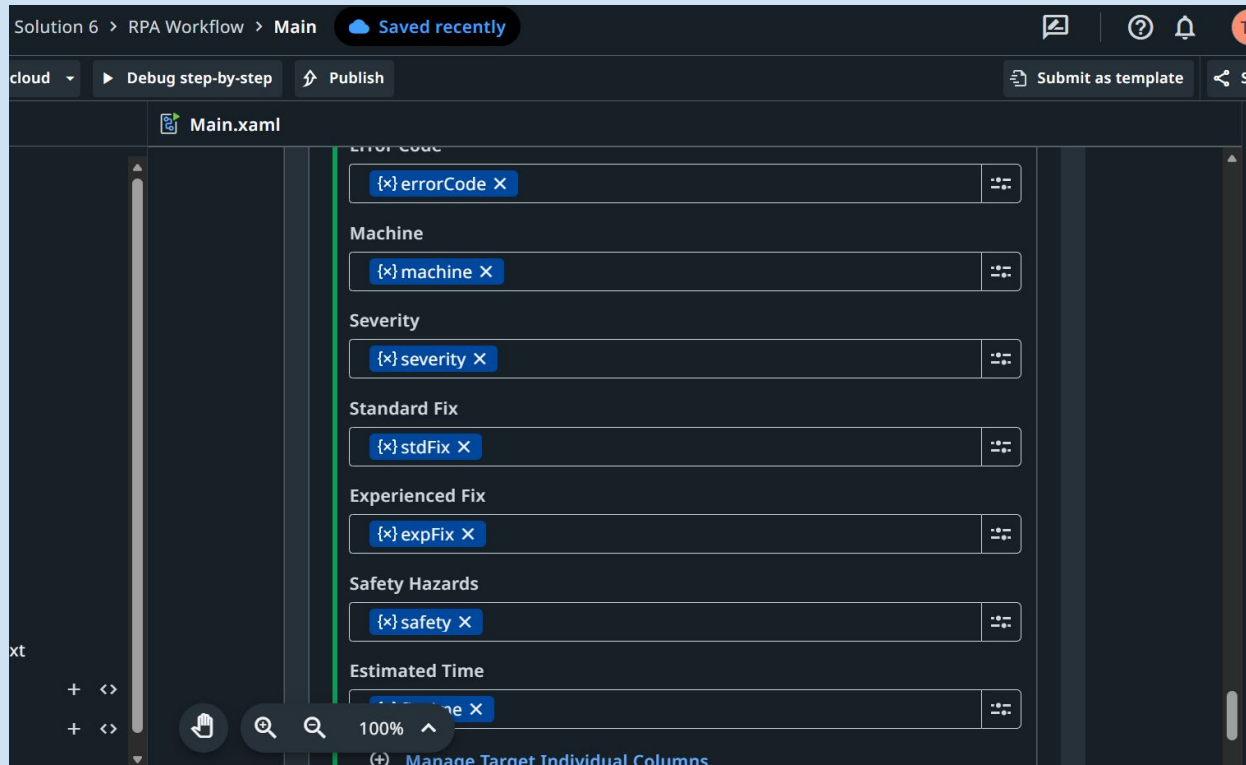
Machine: {x} machine , Error code: {x} errorCode , Severity: {x} severity
Standardfix: {x} stdFix

At the bottom right, there are 'Cancel' and 'Save' buttons. A vertical scrollbar is visible on the right side of the text area.

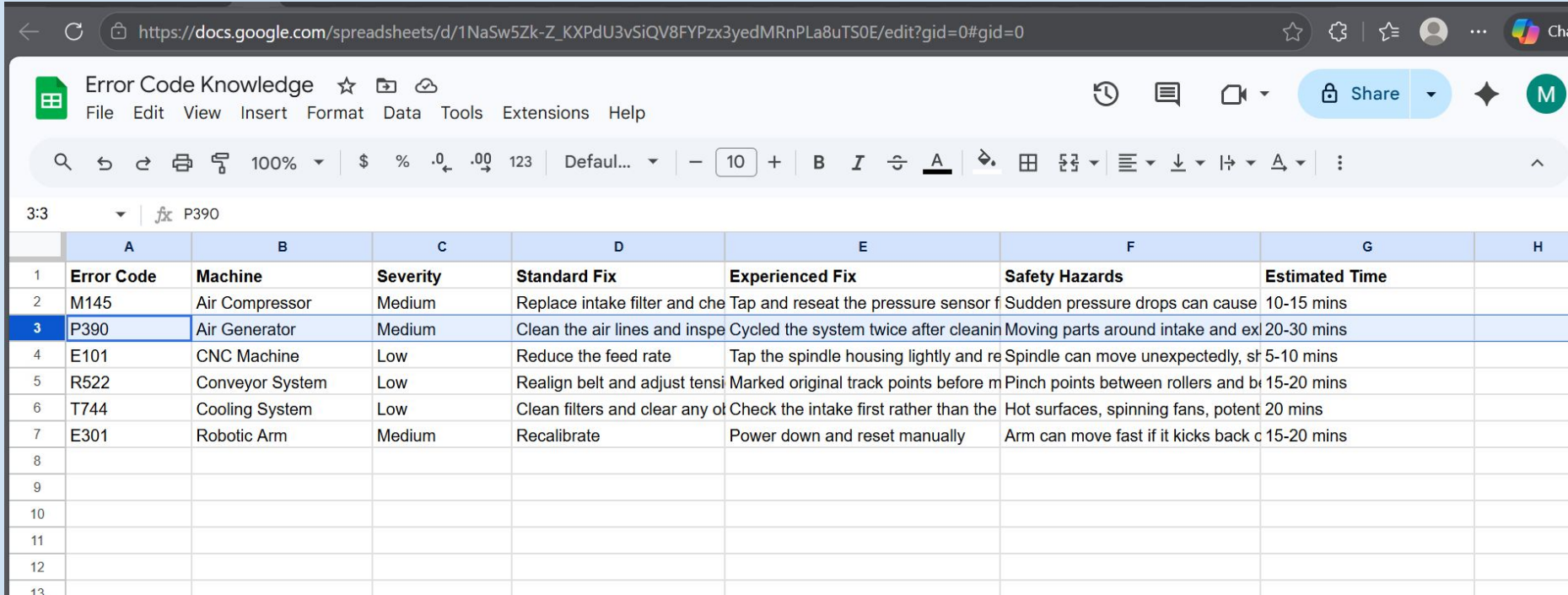
RPA workflow (images:sheet)



RPA workflow (images:sheet)



RPA workflow (images:sheet)



3:3 P390

	A	B	C	D	E	F	G	H
1	Error Code	Machine	Severity	Standard Fix	Experienced Fix	Safety Hazards	Estimated Time	
2	M145	Air Compressor	Medium	Replace intake filter and che	Tap and reseal the pressure sensor f	Sudden pressure drops can cause	10-15 mins	
3	P390	Air Generator	Medium	Clean the air lines and inspe	Cycled the system twice after cleanin	Moving parts around intake and exl	20-30 mins	
4	E101	CNC Machine	Low	Reduce the feed rate	Tap the spindle housing lightly and re	Spindle can move unexpectedly, sh	5-10 mins	
5	R522	Conveyor System	Low	Realign belt and adjust tensi	Marked original track points before m	Pinch points between rollers and b	15-20 mins	
6	T744	Cooling System	Low	Clean filters and clear any of	Check the intake first rather than the	Hot surfaces, spinning fans, potent	20 mins	
7	E301	Robotic Arm	Medium	Recalibrate	Power down and reset manually	Arm can move fast if it kicks back c	15-20 mins	
8								
9								
10								
11								
12								
13								

Benefits

- **Preserves expert knowledge** – Captures workarounds, operational insights, and machine quirks before staff leave or change roles.
 - **Reduces HR effort** – Less time spent gathering knowledge or updating process documents due to non-standard equipment.
- **Standardizes documentation** – Keeps consistent records despite non-standard equipment behavior.
 - **Scalable and repeatable** – Workflow can handle multiple machines or sites, preventing repeated ad hoc fixes.
 - **Supports management oversight** – Critical issues are flagged and sent to management before being shared, ensuring operational control.
- **Reduces downtime** – Technicians can fix rare or patched issues without trial-and-error.
 - **Accelerates onboarding** – New technicians learn real operational methods, not just manuals

Reflection

This workflow highlights the importance of capturing and preserving the knowledge of trusted, experienced technicians. By systematically recording their insights, HR ensures critical operational knowledge isn't lost when staff transition roles or leave. Technicians gain faster access to solutions for rare or non-standard machine issues, reducing downtime and improving safety. Maintenance benefits because non-critical problems are handled by trained staff, freeing them to focus on major repairs. Management retains oversight, reviewing critical issues before dissemination to maintain informed decision-making. Overall, the workflow strengthens organizational resilience, accelerates onboarding, maintains operational consistency, and ensures practical expertise is preserved. By integrating AI and automation, the organization achieves safer, more efficient, and scalable operations while safeguarding invaluable workforce experience.

Challenges & improvements

Challenges and Improvements

Challenges:

- Depends on **trusted, experienced technicians** for accurate insights.
- Knowledge capture occurs **infrequently**, making scheduling critical.
- Ensuring all rare or complex issues are **fully documented**.
- Critical vs. non-critical classification requires **management and maintenance judgment**.
- Variations in responses may affect **consistency and extraction**.

Improvements:

- Integrate knowledge capture into **trusted technician offboarding** or role transitions.
- Adapt workflow for **other areas** that risk loss of experienced knowledge.
- **HR involvement** ensures comprehensive, structured responses.
- **Manager and maintenance review** validates critical insights.
- Knowledge repository is **accessible during the workday** for designated staff or anyone needing guidance.